Udyam Digital

Digital onboarding for Accel Business | implemented web prototype

GEMINI: DRAFTS + EXTRACTION
BACKEND: RULES + DECISIONS

01 Review business needs > **02 Edit catalogue estimate** > **03 Consent + document checks** > **04 Simulated order + tracker**

CUSTOMER / BROWSER

React + TypeScript | responsive web app

Customer workspace

Create account / sign in; English / Hindi
Reviewed needs; editable bundle / quote
Document upload, checkout and tracker
Reviewed chat stays in the open tab

Optional voice input

Separate speech consent + permission
Editable transcript; owner confirms
Transcript becomes advisor description

Audio / transcript

Browser speech service

en-IN / hi-IN recognition
May use an external speech provider
Audio is not stored by this application

JSON / images

FASTAPI BACKEND

Authoritative rules, access checks and workflow

API boundary + account access

Validated schemas, request/rate limits and HttpOnly session cookie
Argon2id password hashes; account ownership and staff roles

Catalogue + quote rules

Select catalogue starter services
Eligibility and product dependencies
Quantities, fees and contract terms

Advisor adapter

Consented text + reviewed facts
Bounded follow-up conversation
Validated draft; guided fallback

Order state machine

Passing checks + accepted terms
Recheck quote; prevent duplicates
Accepted > installation > activation
All service stages are simulated

Document pipeline

PNG/JPEG, max 5 MB; strip metadata
Extract fields; owner reviews
Mock registries, dates, consistency
Unresolved issues block checkout

GEMINI API (EXTERNAL)

Server-side adapter; HTTPS provider calls

Structured assistance

Business text > structured needs draft
Document pixels > extracted fields
Backend validates returned JSON
Provider never sets catalogue prices
Provider never approves checkout

Provider controls

API key stays on the server
Only consented text / pixels are sent
60-second request; one attempt
Advisor outage: guided questions
OCR outage: retry; no guessed fields
Provider data terms apply separately

Read-only catalogue / registry lookups

Save / reopen records

Temporary fields

LOCAL DATA + RETENTION

Different stores for source rules, saved records and temporary fields

Read-only source CSVs

25 products + mock GSTIN / Udyam registries
Deterministic catalogue and registry checks

SQLite: two local databases

accounts.sqlite3: password hashes + hashed sessions
onboarding.sqlite3: consent, summaries, quotes, orders
Journeys 24 h; sessions 8 h; inactive accounts 30 d

Temporary server memory

Extracted fields: up to 30 min; cleared on order submission
Document images are not persisted by the application
Raw fields are lost on server restart

Demo scope: no real payment, availability check, installation or activation. Local SQLite is not encrypted.

Additional web features: consented partner follow-up and authorized review. These are outside the main recorded flow.